

ML Fairness: The Basics

ITAI 2372 | A plain-language introduction

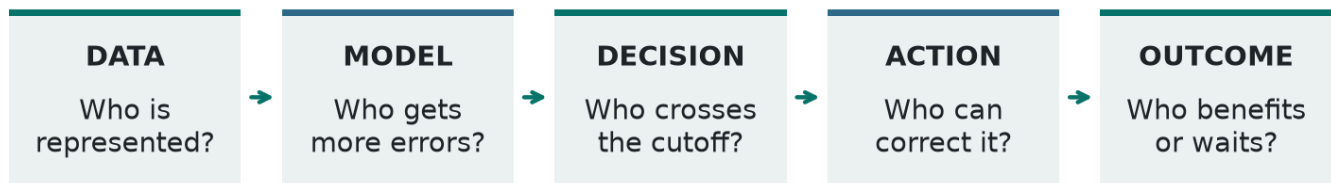
Machine learning (ML) learns patterns from examples to make predictions. **Fairness asks whether the system's use creates unjustified disadvantages for people or groups, and what we should do about them.**

An accurate prediction is not automatically a fair decision. Start with the people affected, not a score on a dashboard. [1]

Two questions to ask

What could go wrong?	Everyday example
Who gets an opportunity?	A qualified person is screened out of a training program.
Who gets a worse service?	A speech tool makes more errors for people with a particular accent.

These are often called **allocation harms** and **quality-of-service harms**. The same system can create both. [1]



Three words worth knowing

- **Bias:** a systematic skew in data, measurement, or decisions. Some forms can contribute to unfair harm, even without anyone intending it.
- **Disparity:** a measured difference between groups, such as one group being missed more often. It is a signal to investigate, not proof of the cause.
- **Proxy:** an indirect stand-in. For example, a location feature may partly reflect income or demographic patterns.

Remember: fairness concerns the whole workflow. A model, a cutoff, and a person's response can each change who benefits or who is harmed.

Same Accuracy, Different Mistakes

Made-up classroom example. These are not results from a real institution.

A training program uses a model to recommend whom to invite. Groups A and B are fictional demographic groups. Each has **20 applicants: 10 eligible and 10 not eligible**, under a stated eligibility rule. Assume those labels are accurate for this exercise.

What the model did	Group A	Group B
Eligible and invited	8	6
Eligible but missed	2	4
Not eligible but invited	2	0
Not eligible and not invited	8	10
Correct decisions, out of all 20	16 / 20 = 80%	16 / 20 = 80%

Now look only at eligible applicants



Each tile = 1 eligible applicant. I = invited M = missed

Group B has twice the missed-eligible rate: 4 out of 10 (40%), compared with 2 out of 10 (20%). Both groups still have 80% overall accuracy because their mistakes are distributed differently.

- A **false negative** here is an eligible person the model does not recommend.
- A **false positive** here is a person recommended despite not meeting the eligibility rule.

The important question is not just "How many mistakes?" It is "Which mistakes, for whom, and with what consequence?"

This small example illustrates the arithmetic; it does not establish a real-world finding. In practice, also check group sizes, uncertainty, and whether the eligibility rule and labels are reasonable. [1, 2]

No Single Fairness Score

Different questions examine different parts of a decision. Choose the question that matches the possible harm, then compare groups. [2]

Question	Who is counted?	A term you may hear
Are similar shares selected?	Everyone in each group	Demographic parity
Are eligible people found at similar rates?	Eligible people in each group	Equal opportunity
Are both kinds of errors similar?	Eligible and ineligible people, checked separately	Equalized odds

In the example, the eligible-person invitation rates are **80% for A and 60% for B**. That is an equal-opportunity gap in this sample. It is not the same as overall accuracy.

These are measurement choices, not universal fairness rules. Criteria can conflict; satisfying one does not settle every concern. An error-rate comparison can also mislead if the labels used as the reference are biased. [2]

Does removing demographic columns fix it?

Not by itself. Other inputs may carry related information. Unequal data coverage, unreliable labels, and the business rule can still produce unequal effects.

Demographic column removed

Other inputs
Historical labels
Decision rule

can still affect



Outcomes for different groups

In our **credit-risk notebook**, removing sex, marital status, and age barely changed AUC. That tells us about **ranking performance**, not whether errors or review decisions are similar across groups. An explanation of a score is not proof of fairness either.

Where appropriate and lawful, a team may use protected group information in a **controlled audit**, separate from model inputs. That requires a valid purpose, privacy safeguards, and authorized access. Legal requirements depend on the context; this handout is not legal advice.

A Simple Fairness Review

Fairness needs repeated checks, with input from affected people, domain experts, technical staff, and the decision owner. [1, 3]

Step	Ask in plain language
1. Name the harm	Who could lose an opportunity or receive worse service?
2. Check the data	Who is missing? Are labels reliable? Check relevant groups and their intersections.
3. Compare results	Who faces more errors or delays? Show group counts and percentages.
4. Investigate and improve	Check the data, model, cutoff, and workflow. Test any changes.
5. Assign responsibility	Who monitors outcomes, corrects mistakes, and handles appeals?

Connect it to complaint routing

Suppose second-language complaints reach human triage more often. **Is that necessarily unfair?** Triage might improve routing, but an understaffed queue could make those consumers wait longer. Check **errors, waiting times, and final service**. A reviewer needs time, information, authority, and oversight. Adding a person is not an automatic fairness fix.

Two-minute class discussion

Return to the training-program example:

1. Which group has the higher rate of missed eligible applicants?
2. Why does equal accuracy not answer the fairness question?
3. What would you investigate before changing the model?

Quick check: B misses 4 / 10 eligible applicants; A misses 2 / 10. Accuracy mixes both error types. Investigate labels, data coverage, group sizes, and the decision rule.

Takeaway: name the harm, compare the relevant groups, investigate the difference, and check what happens after the model's prediction.

Sources and optional reading

- [1] [Fairlearn: Performing a Fairness Assessment](#)
- [2] [Fairlearn: Common Fairness Metrics](#)
- [3] [NIST: AI Risk Management Framework](#) (voluntary guidance)

Sources checked September 11, 2026. All numerical examples in this handout are illustrative.